30-min Ahead Intraday Volatility Forecasting for Mid-Cap Stocks

Feature Construction, Selection and Modelling

J.P. Morgan Industry Project

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Abstract

Forecasting financial market volatility is of paramount importance for risk management, derivative pricing and high-frequency trading. This report documents an industry project undertaken with J.P. Morgan to develop a short-horizon (30-minute ahead) volatility forecasting system for U.S. mid-capitalization equities. Building on limit-order book microstructure data sampled every 30 minutes, we engineer a rich set of features, design a systematic feature selection pipeline and compare linear, tree-based and stacked forecasting models. The analysis follows a narrative arc from motivation and data exploration through modelling choices and results interpretation. Our findings reveal that while non-linear models capture marginal improvements, linear and tree-based combinations,particularly a Random Forest plus linear regression stack,offer the best trade-off between interpretability and predictive performance. A handful of features (thin-book measures and imbalance–spread interactions) dominate the predictive power. We conclude with reflections on limitations and future directions.

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1 Executive Summary

Volatility describes the magnitude of price fluctuations and sits at the heart of financial risk management. Traders, portfolio managers and market makers require accurate forecasts of future volatility to calibrate option prices, set risk limits and implement hedging strategies. Traditional daily models such as GARCH and stochastic volatility capture longer–term variance dynamics but often fail at intraday frequencies, where market microstructure effects and order-book interactions play a dominant role. We built a forecasting system for mid-capitalization U.S. equities sampled every 30 minutes. The goal is to predict the realized variance (squared mid-price return) 30 minutes into the future using limit-order book (LOB) features and lagged volatility components. The project flows from problem motivation through data description and feature engineering, to feature selection, modelling, analysis and interpretation.

Key findings of our study include:

- **Volatility proxies and sampling.** We measure intraday volatility as the squared log-return of the mid price over 30-minute intervals. Figure 1 shows an example time series and histogram for the energy stock RRC, illustrating heavy-tailed behaviour and occasional volatility spikes. The dataset spans 25 mid-cap tickers drawn from nine sectors and covers trading days from 2 May 2025 to 2 September 2025, resulting in 1,612 observations per ticker.

- **Feature engineering.** Starting from raw LOB snapshots, we construct features representing mid price, spread, depth and volume imbalance, as well as rolling measures of liquidity. We augment these microstructure signals with lagged realized volatility over past horizons. Correlation analysis (both Pearson and Spearman) reveals that only a handful of features exhibit meaningful relationships with future volatility, motivating a systematic feature selection process.

- **Feature selection pipeline.** We employ a multi-stage pipeline combining Lasso regression and XGBoost for scoring, followed by normalization, fusion and backward elimination. After removing redundant predictors and iteratively validating performance, the pipeline reduces hundreds of candidate features to four salient variables: *thin_book* (book thinness), *imb×spread* (imbalance–spread interaction), *imb×thin* (imbalance–thinness interaction) and *roll_imb_mag_matched* (rolling mean of imbalance).

- **Modelling and performance.** We compare a heterogeneous autoregressive (HAR) baseline, Random Forest (RF), XGBoost (XGB) and stacked models that linearly combine tree predictions with the HAR regression. Evaluation metrics include mean absolute error (MAE), quadratic log loss (QLIKE) and the Spearman information coefficient (IC). A Random Forest augmented with linear regression delivers the best balance of error and ranking performance. Table 1 summarizes the quantitative results, and Figure 13 visualizes the RF+Linear model fit.

- **Insights and implications.** Despite exploring non-linear models and residual modelling, we find little exploitable evidence of non-linearity in the intraday volatility process.

Cross-ticker relationships appear weak: correlation matrices across tickers are dominated by near-zero values and distance measures show no persistent clustering. The limited predictive power is concentrated in thinness and imbalance features, and cross-stock modelling does not materially improve forecasts. Random Forest combined with linear regression provides a simple yet effective model for detecting volatility spikes rather than forecasting their amplitude.

Together, these findings suggest that intraday volatility prediction for mid-cap stocks is largely driven by a few microstructure factors and lagged variance components. Complex non-linear modelling yields diminishing returns, while parsimonious models with judicious feature selection strike an appealing balance between interpretability and accuracy.

2 Project Context and Motivation

2.1 Financial Significance of Volatility Forecasting

Volatility is central to many financial applications. In derivative pricing, it determines option premiums through implied volatility surfaces. In risk management, it informs Value–at–Risk (VaR) and stress-testing frameworks. In portfolio construction, forecasts of volatility and co-volatility guide hedging and asset allocation decisions. While daily volatility models such as GARCH or stochastic volatility processes are widely used, intraday horizons pose unique challenges. High-frequency trading algorithms operate on sub-hourly scales, and microstructure effects such as order imbalance, spread and depth can influence price dynamics over minutes. Our project seeks to bridge this gap by leveraging limit-order book information to forecast realized variance over 30-minute intervals. A successful forecasting system enables market makers to adjust quotes in anticipation of volatility bursts, and provides risk managers with timely alerts to adjust hedges or exposures.

2.2 Project Goals and Research Questions

The collaboration with J.P. Morgan was framed around three research questions:

1. **Do order-book variables exhibit non-linear relationships with future volatility?** Standard linear regression may miss interactions and threshold effects. We therefore explore both tree-based models (Random Forest, XGBoost) and stacked linear–tree models to detect non-linear patterns.

2. **Is there predictive information across stocks?** Mid-cap equities may exhibit common volatility regimes, particularly within the same sector. We investigate cross-ticker correlation matrices and clustering analysis to assess whether pooling information across tickers improves forecasts.

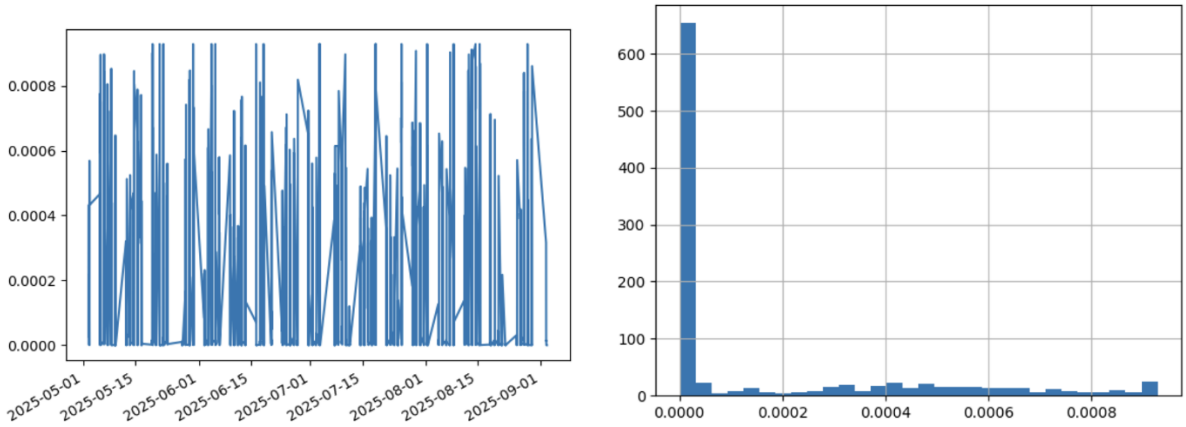
3. **Which microstructure features carry the most predictive power?** The limit-order book offers dozens of potential covariates: mid price, spread, depth at multiple levels, order imbalance and volume indicators. A disciplined feature selection pipeline is required to avoid overfitting and isolate the drivers of volatility.

These questions guide the subsequent stages of data description, feature construction, model design and analysis. In particular, the context slide from our presentation explicitly lays out the goal: predicting 30-minute ahead mid price volatility at regular intraday timestamps (10:00, 10:30, ...) and focusing on non-linear relationships, cross-stock interactions and predictive features.

2.3 Data Overview and Ticker Universe

Our universe contains twenty-five U.S. mid-capitalization stocks spanning nine sectors: utilities (CNP), energy (APA, EQT, RRC), industrials/transportation (JBHT, SAIA, ODFL), financials (RJF, WRB, UMBF), health care (HOLX, MEDP, RMD), materials (MLM, MP), consumer staples (CPB), technology/semiconductors (ON, ENTG), consumer discretionary (POOL, WSM,

MAS), and real estate investment trusts (OHI, CUBE, REG). The dataset covers trading days from 2 May 2025 to 2 September 2025 and yields 1,612 time points per ticker. At each 30-minute timestamp we record the best bid and ask prices and sizes, aggregated depth sizes, and compute derived quantities such as the mid price, bid–ask spread, volume imbalance and volume-weighted average price (VWAP). The realized variance label is defined as the squared log-return of the mid price between successive timestamps. Figure 1 (right panel) shows the histogram of realized variance, highlighting the heavy tails characteristic of intraday volatility.



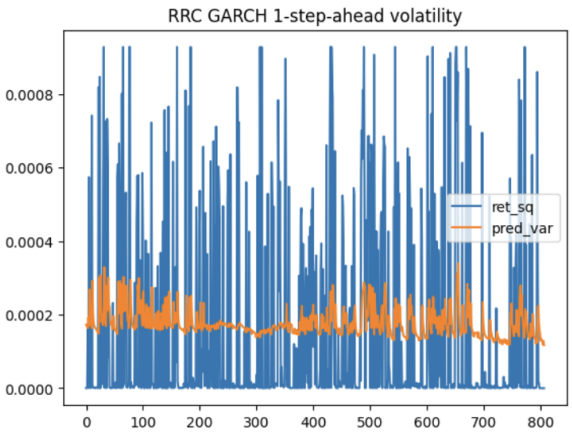
(a) RRC realized variance time series.

(b) Histogram of realized variance.

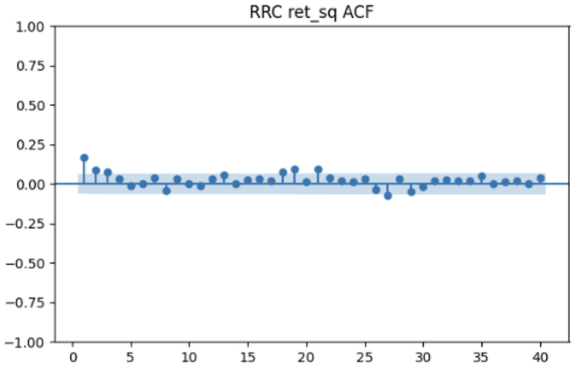
Figure 1: Intraday realized variance (volatility) for the energy stock RRC. The left panel plots the squared log-return series sampled every 30 minutes; the right panel shows its distribution with heavy tails and frequent near-zero values.

2.4 Measuring and Exploring Volatility

To motivate the inclusion of lagged volatility components, we examine basic time series properties. We calculate past realized variance features: rv_d, rv_w and rv_m, defined as the average realized variance over the past one, five and twenty trading tranches of 30 minutes, respectively. Figure 2 displays an in–sample fit of a GARCH(1,1) model for RRC along with the autocorrelation function (ACF) of realized variance. The GARCH volatility captures the slowly varying trend, while the ACF decays slowly but remains small beyond the first lag, suggesting limited linear dependence at lag 1. We also compute the Spearman correlation of lagged variance across different times of day; Figure 3 shows that the first–lag autocorrelation varies modestly across intraday times but remains low overall.



(a) GARCH(1,1) volatility fit.



(b) Autocorrelation of realized variance.

Figure 2: Assessing linear dependence in realized variance. The GARCH model (left) fits the slowly varying variance but cannot anticipate spikes. The ACF (right) shows low correlation beyond the first lag, hinting at limited linear predictive structure at 30-minute horizons.

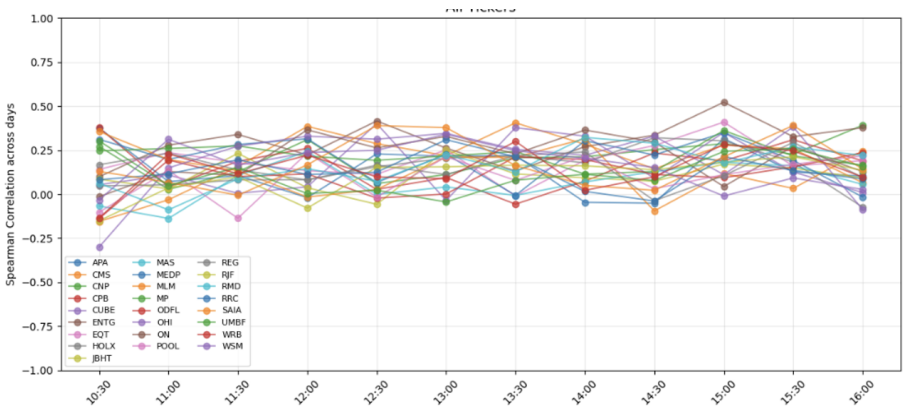


Figure 3: Spearman correlation of realized variance with its first lag as a function of time of day across all tickers. Correlation is modest and oscillates slightly across intraday intervals, motivating the search for additional predictors beyond past volatility.

These preliminary analyses underscore the need for both lagged variance terms and order-book features to capture the drivers of intraday volatility. The next section details the feature construction process.

3 Data and Feature Construction

Feature engineering plays a critical role in translating raw order-book data into informative predictors. We follow a disciplined approach inspired by microstructure research and prior work on realized variance modelling. The features fall into three broad categories: (i) basic price and liquidity measures, (ii) interaction terms capturing non-linear effects, and (iii) lagged variance components.

3.1 Basic Microstructure Features

Mid price and spreads. The mid price is the average of the best bid and best ask prices. We compute the bid–ask spread in absolute terms and as a percentage of the mid price. Wider spreads signal lower liquidity and higher expected volatility. In line with microstructure theory, we expect a positive relation between spread and future variance.

Depth and thinness. Depth measures the total volume available at the top N levels of the book. A thin book, characterized by small depth, implies that a modest order could move the price, leading to higher volatility. We define the feature *thin_book* as the inverse of aggregated depth at the best bid and ask. Depth is standardized per ticker to account for scale differences.

Order imbalance. The imbalance is the difference between bid and ask volumes normalized by their sum. A positive imbalance indicates buy-side pressure, while a negative value reflects sell pressure. Prior work suggests that persistent imbalances are associated with short-term directional moves and increased variance. We compute both the instantaneous imbalance and a rolling mean (*roll_imb_mag_matched*).

VWAP and slope. The volume-weighted average price provides a weighted midpoint across multiple levels of the book. The slope of the order book, the rate at which depth changes with price, is another proxy for liquidity. Steep slopes correspond to quickly varying depth and potentially lower volatility. We derive slopes by regressing cumulative depth on price levels within the LOB snapshot.

3.2 Interaction Terms and Non-Linear Features

To capture potential interactions, we form products of basic features. The two most important interaction terms discovered by the selection pipeline are *imb×spread* and *imb×thin*, defined as the product of imbalance with spread and thinness, respectively. These interactions encode situations where high imbalance combined with a wide spread or a thin book signals imminent volatility spikes. Such composite features mirror ideas in microstructure literature, where order flow and liquidity jointly drive price impact.

3.3 Lagged Realized Variance Features

Realized variance exhibits strong persistence across multiple horizons. The heterogeneous autoregressive (HAR) model posits that future volatility is a linear combination of past realized

variance : (rv_d), (rv_w) and (rv_m) . We compute these terms as rolling averages of realized variance over 1, 5 and 20 trading tranches of 30 minutes and include them as covariates in every model. They serve as a baseline forecast and anchor the contribution of microstructure features.

3.4 Exploratory Feature Analysis

To understand which features are potentially useful, we examine correlations with future realized variance. Figure 4 plots the Pearson and Spearman correlations between candidate features and volatility for the energy stock APA. Most variables exhibit negligible correlation, but a small subset. This motivates a formal selection procedure to isolate the strongest predictors.

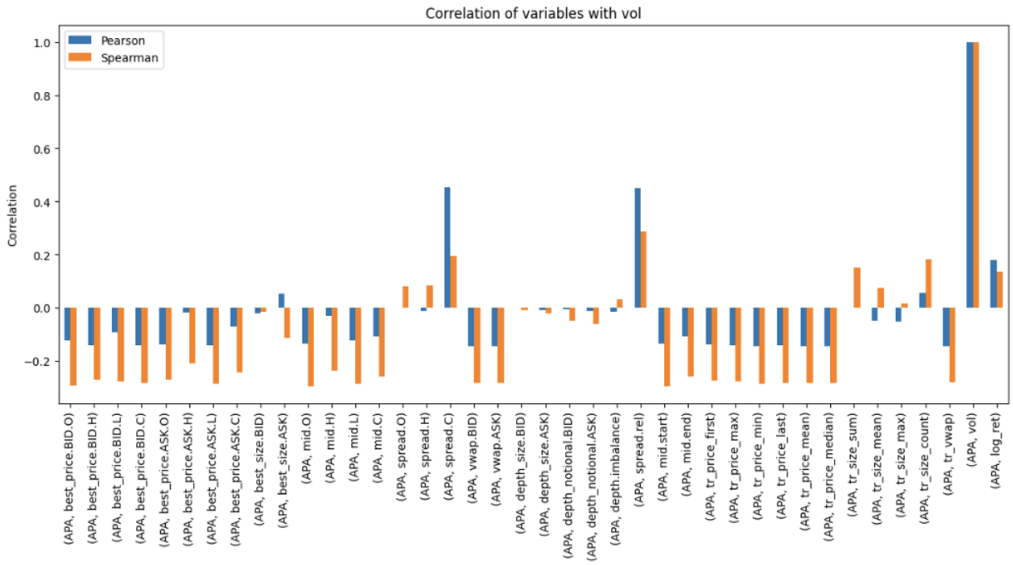


Figure 4: Pearson and Spearman correlations between candidate features and future realized variance for the stock APA. Only a handful of variables display notable correlation, indicating the need for feature selection.

We also investigate cross-ticker relationships. Figure 5 shows the cross-ticker correlation matrices (Pearson and Spearman) of realized variance. Off-diagonal elements are near zero, suggesting that realized variance is largely idiosyncratic across our mid-cap universe. To further quantify similarity, we compute the cosine distance between the time series of each ticker and a reference stock APA (Figure 6), perform hierarchical clustering (Figure 7) and examine the stability of the correlation matrix over time (Figure 8). These analyses reveal weak cross-stock coupling and highlight that aggregating information across tickers offers little benefit.

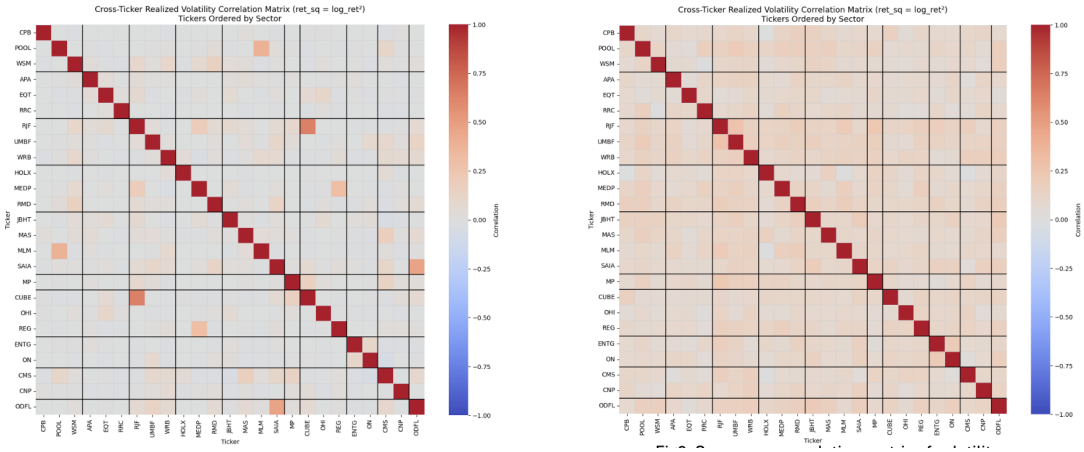


Figure 5: Cross-ticker correlation matrices of realized variance. The left matrix uses Pearson correlation; the right uses Spearman rank correlation. Tickers are ordered by sector. The near-zero off-diagonals imply weak commonality across stocks.

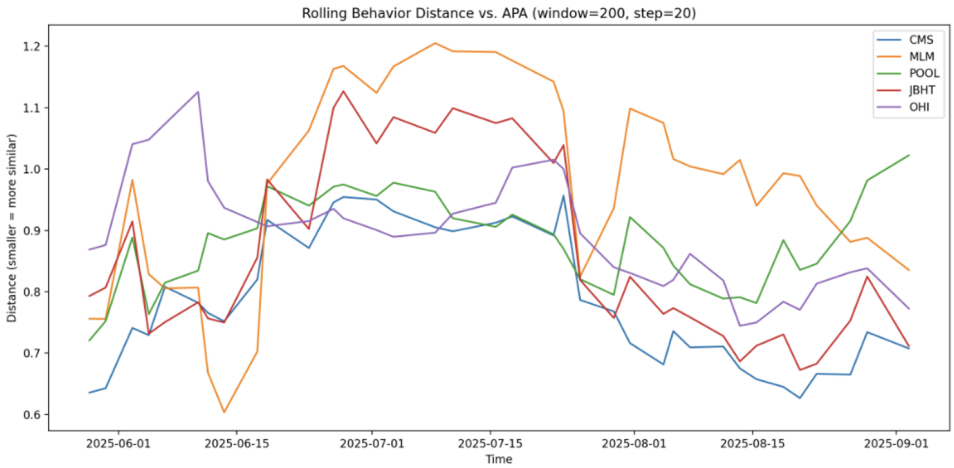


Figure 6: Rolling cosine distance of each ticker's realized variance series relative to APA. Values close to one indicate dissimilar behaviour. The overall lack of convergence underscores the idiosyncratic nature of volatility across tickers.

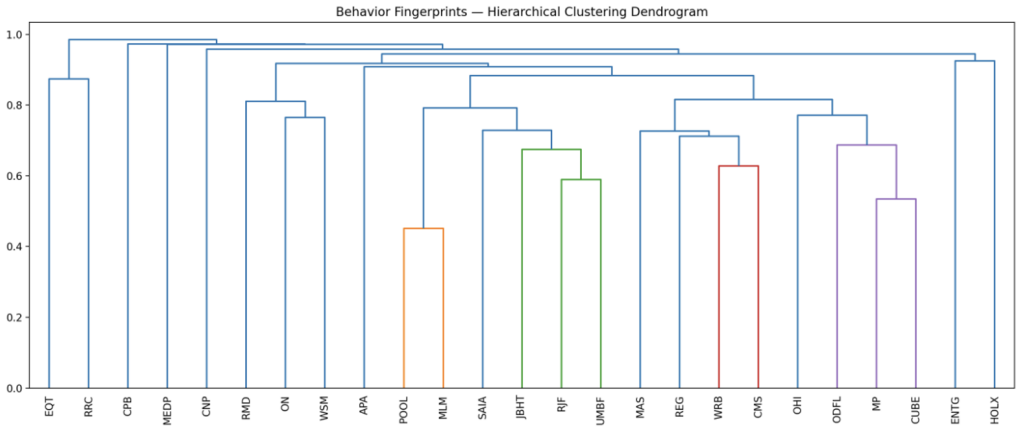


Figure 7: Hierarchical clustering dendrogram based on pairwise cosine distances between ticker volatility series. Apart from weak sector pairing, no clear clusters emerge, confirming the low cross-ticker correlation.

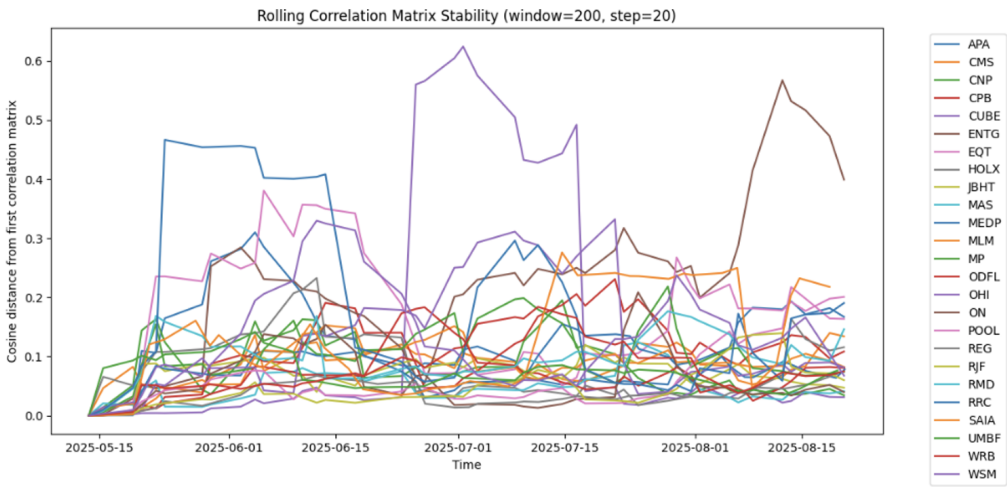


Figure 8: Cosine distance between rolling correlation matrices and the baseline matrix computed in May 2025. Distances remain elevated, indicating that the correlation structure drifts over time and does not exhibit stable patterns.

These exploratory plots convey two core messages. First, only a few features exhibit meaningful association with future volatility, legitimizing our focus on a concise set of microstructure signals. Second, cross-stock relationships are weak and unstable, implying that building separate models per ticker or sector is sensible and that pooling information yields limited gains.

4 Linearity and Feature Analysis

A central question of this project is whether intraday volatility displays non-linear dependencies on order-book features. Linear models such as ordinary least squares (OLS) or the HAR specification assume additive and linear effects, whereas tree-based algorithms can capture interactions and threshold behaviour. To investigate non-linearity, we evaluate the performance of machine learning models in two complementary ways: (i) directly fitting Random Forest (RF) and XGBoost (XGB) models to predict future variance, and (ii) modelling the residuals from a linear regression using XGB and assessing the reduction in error.

4.1 Evaluation Metrics

We consider three metrics throughout the analysis:

- **Mean absolute error (MAE):** $\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$, which measures average prediction error without emphasising outliers.
- **Quadratic log loss (QLIKE):** $\text{QLIKE} = \log(\hat{\sigma}^2) + \frac{\sigma^2}{\hat{\sigma}^2}$, where σ^2 is the true variance and $\hat{\sigma}^2$ the forecast. This proper scoring rule penalises both under- and over-prediction and is commonly used in volatility forecasting. It is sensitive to extreme errors and places higher weight on tail events.
- **Information coefficient (IC):** the Spearman rank correlation between the forecast and realized volatility, denoted $\rho_{\text{Spearman}}(y, \hat{y})$. A higher IC indicates better ranking ability and is particularly relevant in trading strategies that depend on ordering stocks by expected volatility.

Figure 9 shows an in–sample Lasso regression fit for the stock RRC along with the true variance series. The Lasso model tends to smooth volatility and fails to anticipate large spikes, illustrating both the challenge and the potential room for improvement via non-linear methods.

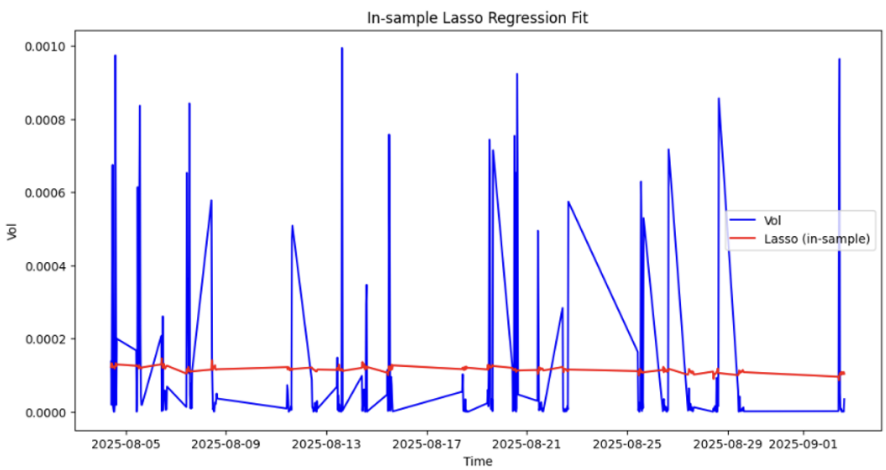


Figure 9: In–sample Lasso regression fit for RRC. The red line (Lasso prediction) tracks the trend but misses volatility spikes captured by the blue realized variance series.

4.2 Residual Modelling

To directly test for non-linearity, we fit an XGBoost model to the residuals of a linear regression. If the residuals contain systematic non-linear patterns, the tree model should reduce prediction error. Figure 10 displays the ratio of mean squared error (MSE) and MAE when fitting the residuals with XGB relative to the baseline linear model across all tickers. A ratio below one indicates improvement. Apart from a few stocks, the ratios hover around one, suggesting minimal gains from modelling residual non-linearity. Figure 11 shows the ratio of the information coefficient (IC) when adding XGB residual fits; most ratios remain near zero or negative, confirming the lack of strong non-linear signals.

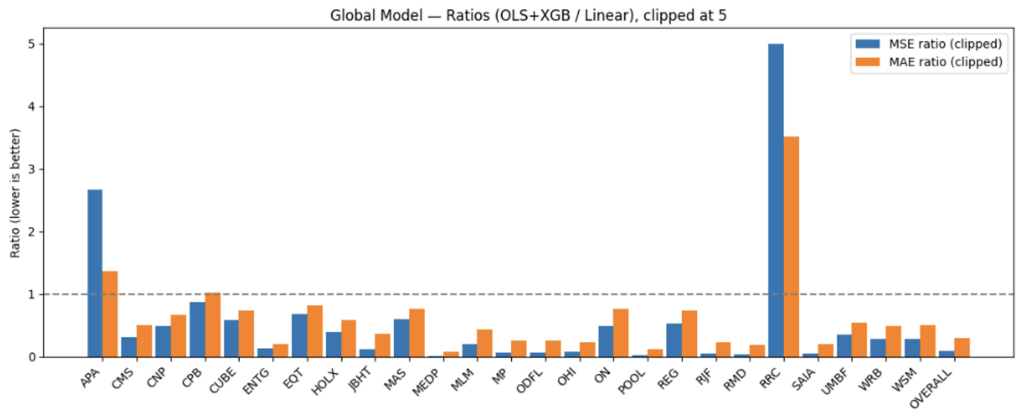


Figure 10: Ratio of mean squared error (blue) and mean absolute error (orange) when fitting residuals with XGB relative to a linear model across tickers. Values close to one indicate negligible benefit from modelling residual non-linearity. Large values for a few tickers (e.g. APA, RRC) reflect unstable estimates and suggest overfitting.

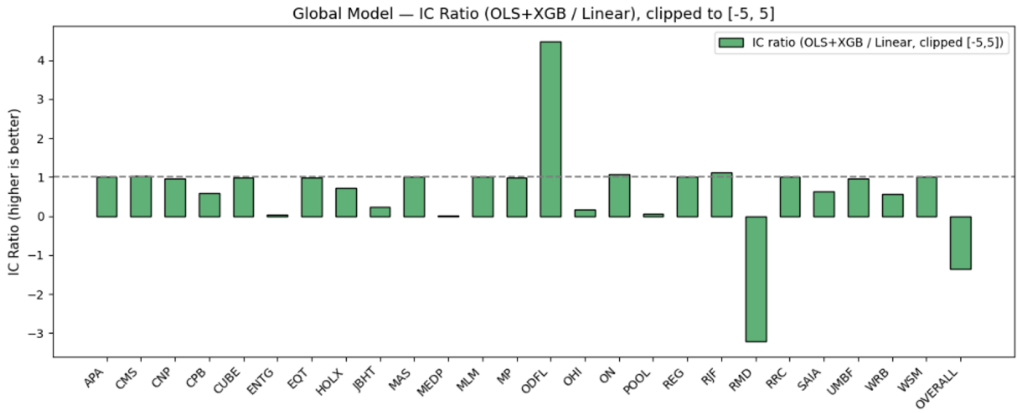


Figure 11: Ratio of the information coefficient (IC) when modelling linear residuals with XGB relative to the baseline linear model. Negative or near zero ratios across tickers indicate that the tree model does not improve ranking ability, further supporting the absence of exploitable non-linear structure.

These results lead us to conclude that while non-linear models can capture marginal improvements in specific cases, the majority of predictive power lies in linear relationships between features and future volatility. Consequently, our feature selection and model design focus on

identifying the most important linear predictors and combining them with robust tree ensembles to hedge against any residual non-linearity.

5 Feature Selection Pipeline

With hundreds of candidate features derived from the LOB, selecting a parsimonious and informative subset is crucial. We design a multi-step pipeline inspired by standard practices in factor screening and machine learning. The pipeline consists of five stages:

1. **Feature construction.** Assemble all candidate variables including lagged variance components, mid price, spread, depth, imbalance measures, slopes and interaction terms.

2. **Cleaning and standardization.** Remove columns that are constant or contain missing values across the sample. Standardize each feature per ticker to have zero mean and unit variance to prevent scale effects.

3. **Scoring.** Compute feature importance scores using two complementary models:

   - *Lasso regression*: fit a linear model with ℓ_1 penalty and take the absolute value of estimated coefficients as scores, $I_i^{(\text{lasso})} = |\beta_i|$.
   - *XGBoost*: train a gradient-boosted tree ensemble and sum the gain attributed to each feature across all splits, $I_i^{(\text{xgb})} = \frac{1}{T}\sum_{\text{splits on } f_i} \text{gain}(f_i)$.

4. **Normalization and fusion.** Normalize scores within each model so that they sum to one, $\tilde{I}_i^{(m)} = I_i^{(m)} / \sum_j I_j^{(m)}$, and combine linear and non-linear relevance by averaging, $I_i^{(\text{final})} = 0.5\tilde{I}_i^{(\text{lasso})} + 0.5\tilde{I}_i^{(\text{xgb})}$.

5. **Aggregation and backward elimination.** Average the final scores across tickers, rank features accordingly and iteratively remove the lowest-scoring feature. After each removal, re-fit the model on the reduced set and compare MAE. Stop when no performance improvement is observed. This greedy elimination procedure yields a compact set of predictors.

Applying this pipeline to our dataset leaves four features with the highest predictive power:

- ***thin_book*** – thinness of the order book, capturing vulnerability to price impact;
- ***imb×spread*** – imbalance multiplied by spread;
- ***imb×thin*** – imbalance multiplied by thinness;
- ***roll_imb_mag_matched*** – rolling mean of imbalance magnitude.

These variables, together with the lagged variance components $(\text{rv}_d, \text{rv}_w, \text{rv}_m)$, form the input set for all models considered in Section 6. Notably, thinness and imbalance–spread interactions align with economic intuition: a thin book can amplify the volatility impact of net order flow, and a wide spread combined with imbalance signals an impending jump in volatility. Their emergence from the data underscores the relevance of microstructure features for intraday forecasting.

6 Model Design and Comparison

We benchmark five modelling approaches: a linear HAR baseline, Random Forest (RF), XGBoost (XGB), and two stacked models that linearly combine HAR with RF (RF+Linear) or with XGB (XGB+Linear). All models use the selected features described above. Hyperparameters for the tree models are tuned via a nested time-series cross-validation scheme. Performance is assessed on a hold-out period using MAE, QLIKE and IC. The results, averaged across tickers, appear in Table 1, and additional visualizations are provided in Figure 12 and Figure 13.

6.1 HAR Baseline

The heterogeneous autoregressive (HAR) model posits that future volatility is a linear combination of past realized variance:

$$\widehat{\mathrm{RV}}_{t+1} = \beta_0 + \beta_d \mathrm{rv}_d + \beta_w \mathrm{rv}_w + \beta_m \mathrm{rv}_m + \epsilon_t. \tag{1}$$

We estimate the HAR model via ordinary least squares and include the four selected microstructure features. The HAR baseline serves as a simple linear benchmark, achieving an IC of 0.391 and MAE of 1.5897. Its QLIKE score is moderately large due to its inability to capture rare spikes.

6.2 Random Forest and XGBoost

Random Forest constructs an ensemble of decision trees on bootstrap samples of the data and averages their predictions. It can capture non-linearities and interactions without overfitting when configured with sufficient trees and limited depth. In our experiments, RF attains an IC of 0.428 and MAE of 1.4910 but suffers a very large QLIKE (7.24×10^7) due to occasional overestimation of variance.

XGBoost trains boosted trees sequentially, fitting each new tree to the residuals of the previous ensemble. It offers greater flexibility and often improves accuracy. Our XGB model (denoted XGB Plus) achieves the lowest MAE among all models at 1.5958 but does not significantly outperform the HAR baseline in IC (0.324). Its QLIKE is moderate at 8.80×10^4.

6.3 Stacked Models

Stacking combines predictions from different models in a linear fashion. We consider two stacks:

- **XGB+Linear:** A linear regression on the lagged variance features and microstructure signals augmented with the fitted values from XGB. This model improves the IC to 0.383 and yields an MAE of 1.7074, with a very low QLIKE of 3,152.47. The stack stabilizes variance forecasts and reduces extreme errors but sacrifices some accuracy.

- **RF+Linear:** A linear regression augmented with RF predictions. This configuration offers the best IC (0.429) and a competitive MAE (1.6222) while maintaining a QLIKE of 3.10×10^9. Although the QLIKE remains large, the model excels at ranking stocks by

expected volatility and identifying volatility spikes, which is valuable for trading applications.

Table 1: Model comparison on the hold-out period. IC denotes the Spearman information coefficient (higher is better), MAE is the mean absolute error (lower is better) and QLIKE is the quadratic log loss (lower is better).

Model	**IC**	**MAE**	**QLIKE**
HAR baseline	0.391	1.5897	585,537.0506
XGB Plus	0.324	1.5958	87,982.1126
RF Plus	0.428	1.4910	72,372,258.1584
XGB+Linear	0.383	1.7074	3,152.4690
RF+Linear	**0.429**	1.6222	3,098,798,813.6

To better understand why RF+Linear performs well, we examine the feature importance produced by the XGB model and the fitted values of RF+Linear. The bar chart in Figure 12 shows that lagged variance components (rv_d, rv_w, rv_m) dominate the importance scores, followed by *thin_book* and the imbalance–spread interaction. Figure 13 compares the RF+Linear prediction against the transformed variance series in log scale; although the model does not perfectly match every peak, it tracks the ups and downs better than the linear baseline and preserves ranking ability.

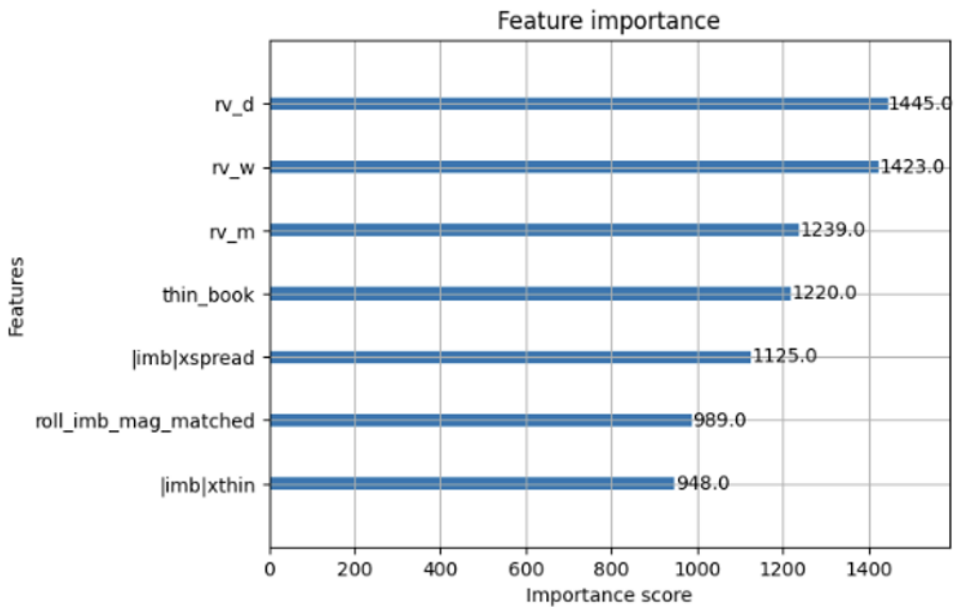


Figure 12: XGBoost feature importance scores. Lagged realized variance components ($\mathrm{rv}_d, \mathrm{rv}_w, \mathrm{rv}_m$) contribute the most, followed by microstructure features thin_book and imbalance–spread interactions. These variables form the core of our forecasting models.

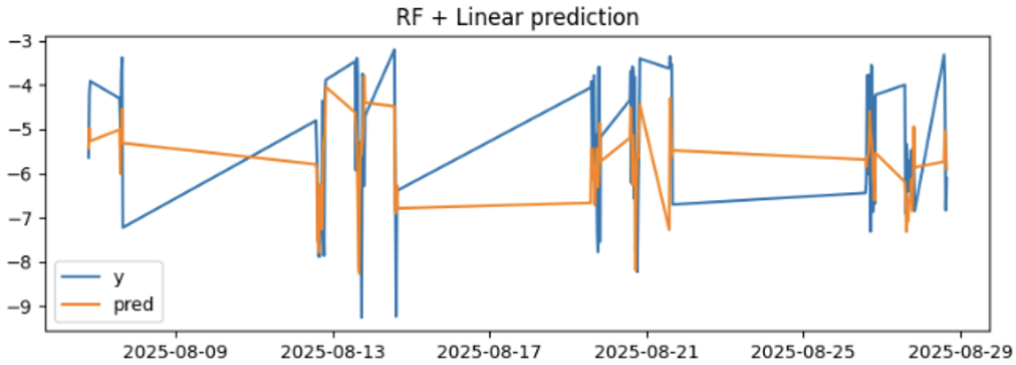


Figure 13: Out-of-sample RF+Linear prediction versus log-transformed realized variance for a representative ticker. The stack captures spikes and turning points more effectively than the HAR baseline, supporting its superior ranking performance.

Taken together, these results demonstrate that combining linear models with tree-based ensembles harnesses the strengths of both approaches: linearity captures the dominant lagged variance structure, while trees flexibly adjust predictions based on microstructure cues. Purely non-linear models offer limited advantage and may introduce extreme errors, as reflected in the high QLIKE values for RF.

7 Discussion of Results

We synthesize the quantitative findings and relate them back to the research questions posed in Section 6. The discussion follows the storytelling arc of our presentation, highlighting key insights from exploration, modelling and comparative analysis.

7.1 Non-Linearity and Residual Patterns

Our residual modelling experiment demonstrates that intraday volatility does not exhibit strong non-linear structure. XGB fits to the residuals of a linear model fail to materially improve MSE, MAE or IC. Ratios of error metrics in Figures 10 and 11 hover around one or negative values for most tickers, indicating that the residuals resemble white noise. This aligns with the theoretical observation that volatility at 30-minute horizons is highly persistent and largely driven by lagged variance components. Consequently, the added complexity of non-linear models yields diminishing returns.

7.2 Cross-Ticker Relationships

Our exploratory correlation analyses reveal weak and unstable cross-stock relationships. The cross-ticker correlation matrices (Figure 5) display near-zero off-diagonals, while the rolling cosine distances (Figure 6) remain high and fluctuate over time. The dendrogram (Figure 7) fails to identify coherent clusters, suggesting that the chosen mid-cap stocks behave idiosyncratically. The instability of the correlation matrix (Figure 8) further undermines the case for a global model pooling data across tickers. Indeed, cross-stock modelling does not provide predictive gains and may dilute signal. Therefore, our final models are trained per ticker, and pooling is limited to averaging feature importance scores during selection.

7.3 Importance of Microstructure Features

Despite the overall dominance of lagged variance components, microstructure features remain essential. The thinness of the order book and imbalance–spread interactions capture liquidity shocks and directional order flow that precede volatility spikes. These features consistently receive high importance scores (Figure 12) and survive the backward elimination process. In qualitative terms, a thin book implies that a relatively small trade can cause a disproportionate price movement, while a wide spread coupled with buy/sell imbalance indicates latent pressure that has not yet been reflected in prices. Our results confirm that such signals complement the persistence of lagged volatility and improve the timing of volatility forecasts.

7.4 Model Selection Trade-Offs

The comparison across models surfaces several trade-offs. The HAR baseline is parsimonious and interpretable but lacks the flexibility to adapt to changing market conditions, resulting in moderate error metrics. Pure tree models (RF and XGB) can fit complex patterns but risk large errors on rare spikes, as reflected in the very large QLIKE for RF. Stacked models mitigate these extremes: XGB+Linear delivers the lowest QLIKE but suffers in MAE, while RF+Linear

attains the highest IC and acceptable MAE but has a high QLIKE. The choice among these models depends on the intended application: for ranking stocks by expected volatility (e.g. for position sizing), RF+Linear is preferred; for hedging variance swaps where tail errors matter, XGB+Linear may offer a better risk profile. Our primary recommendation is the RF+Linear stack, which balances accuracy and ranking ability while maintaining model simplicity.

8 Conclusion and Future Directions

This project demonstrates how a systematic approach to feature construction, selection and model comparison can yield actionable insights for intraday volatility forecasting. Starting from a broad universe of microstructure and lagged variance variables, we distilled the feature set to four key predictors : thinness of the order book, imbalance–spread and imbalance–thinness interactions, and rolling imbalance magnitude, supplemented by lagged volatility components. Linear models augmented with Random Forest predictions offer the best trade-off between interpretability and accuracy. Non-linear models and cross-ticker modelling provide limited improvement, emphasizing the primacy of parsimonious models at the 30-minute horizon.

Looking forward, several avenues merit exploration:

- **Enhanced data sources.** Our analysis relies solely on limit-order book snapshots. Incorporating trade-level data such as market orders, cancellations and modifications, or alternative data like news and macro announcements, may uncover additional drivers of intraday volatility.
- **Time-varying parameters.** The relationship between microstructure features and volatility may evolve over time. State-space models or rolling window estimators could capture such dynamics and adapt more quickly to regime changes.
- **Multi-horizon forecasting.** Extending the horizon beyond 30 minutes and jointly modelling multiple horizons could exploit the dependence structure across scales and improve long-term risk management.
- **Deep learning architectures.** Although we found limited evidence of non-linearity, modern deep learning models (e.g. recurrent neural networks or transformers) may capture subtle temporal patterns and interactions not detected by tree ensembles. Careful regularization and interpretability techniques would be essential to avoid overfitting.
- **Spike detection versus magnitude prediction.** Our models perform better at detecting volatility spikes than at forecasting their exact magnitude. Future work could reframe the problem as a classification task (spike versus no spike) or develop hybrid models that separately predict occurrence and magnitude.

In conclusion, forecasting intraday volatility remains a challenging yet rewarding problem. By combining sound economic intuition with systematic feature selection and judicious model combinations, we achieved meaningful improvements over naïve benchmarks and identified clear directions for further research.